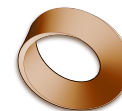


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



1.6 – The EM Spectrum

Unit 1: Light & the Electromagnetic Spectrum

Learning intentions

- a) I can list the seven regions of the electromagnetic spectrum in order of increasing energy.
- b) I can explain that visible light is just a small part of a much larger family of waves.
- c) I can use a mnemonic to remember the correct order of the spectrum regions.

Theory

We often think of light as only what we can see, such as the colours of a rainbow. However, **visible light** is actually just one tiny slice of a much bigger family of waves called the **electromagnetic spectrum**. This spectrum includes all types of electromagnetic radiation, which are waves that carry energy through space. While our eyes can detect visible light, other parts of this family are invisible to us but can be detected by special instruments or felt as heat.

The regions of the spectrum are arranged based on their energy levels. The energy increases as you move from one end of the spectrum to the other. At the low-energy end, we have **radio waves**, which are used for broadcasting music and television signals. As the energy increases, we encounter **microwaves** (used in cooking and mobile phones), followed by **infrared** radiation, which we experience as heat from the Sun or a warm radiator.

Moving further up the energy scale, we reach **visible light**, then **ultraviolet** rays (which can cause sunburn), **X-rays** (used by doctors to look at bones), and finally **gamma rays**, which have the highest energy of all and come from radioactive materials and space events. To help remember this order, scientists use a **mnemonic** — a memory aid that helps us recall lists or sequences easily.

Word bank: gamma · radio · infrared · ultraviolet · visible light · microwaves · X-rays

Activity

Create your own mnemonic to remember the order of the electromagnetic spectrum from lowest energy (radio waves) to highest energy (gamma rays). You can use the first letter of each region: **R, M, I, V, U, X, G**.

- Write down the seven regions in order below, using your mnemonic to help you.
- Draw a simple arrow or ladder next to them and label it "Increasing Energy".

Region Name	First Letter	Mnemonic Word/Phrase (Start with this letter)
1. _____	R	
2. _____	M	
3. _____	I	
4. _____	V	
5. _____	U	
6. _____	X	
7. _____	G	

*Example Mnemonic: **R**abbits **M**ake **I**ce **V**ery **U**nusual **X**-mas **G**ifts.*

Questions

1. Which part of the electromagnetic spectrum can human eyes detect directly?

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2. Name the region of the spectrum that has **more** energy than visible light but **less** energy than X-rays.

3. True or False: Radio waves have higher energy than gamma rays. Explain your answer.

4. Why is it useful for scientists to use a mnemonic when studying the electromagnetic spectrum?

Extension

Find out one specific real-world application for **microwaves** and one for **ultraviolet** radiation. How do these uses relate to the properties of those specific waves?
